
Vehicle Market Analytics System

Normalization Analysis — 1NF, 2NF, and 3NF

Milestone 2 — Database Normalization & Final Schema

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Normalization Summary

12	3	3NF
Tables	NF Levels	Final Normal Form

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1. Introduction

This document covers Milestone 2 of the Vehicle Market Analytics System: normalization of the database schema from a raw flat CSV to Third Normal Form (3NF), followed by the final schema for all 12 tables. The source data was a single flat PakWheels CSV with 18 columns and 497 records. Normalization eliminates redundancy, removes partial and transitive dependencies, and produces 12 well-defined relational tables.

2. First Normal Form (1NF)

Rule: Every column holds a single atomic value; no repeating groups; each row is uniquely identified by a primary key.

Table	Issue / Status	Change Made & Reason
<code>raw_vehicles.im</code>	Multi-valued: only one image URL stored per row; multiple images per listing not representable.	Extracted to <code>vehicle_images</code> table — one row per image with <code>vehicle_id</code> FK. Repeating groups violate 1NF; multi-valued attributes require a separate relation.
<code>raw_vehicles.ci</code>	Non-atomic: city and province sometimes encoded in a single string (e.g., "Lahore, Punjab").	Split into atomic <code>city</code> and <code>state</code> columns in a dedicated <code>locations</code> table. Each column must store one indivisible value.
<code>raw_vehicles.id</code>	Scraped ID is not a guaranteed unique or stable primary key.	Replaced with <code>AUTO_INCREMENT</code> surrogate PKs on all tables. Every relation requires a reliable unique row identifier.
<code>price_currency</code>	Constant value 'PKR' repeated across all 497 rows — zero information value.	Column removed; currency is fixed at the application layer. A constant column wastes storage and adds no relational information.
<code>detail_url, scraped_at</code>	ETL scraping metadata stored alongside business attributes.	Removed from the business table; retained in the ETL pipeline context only. Business and ETL concerns must not share a table.
All remaining columns	Values are atomic; each row is distinct; no repeating groups exist.	No change required. 1NF is already satisfied.

3. Second Normal Form (2NF)

Rule: The table is in 1NF and every non-key attribute is fully functionally dependent on the entire primary key. With single-column surrogate PKs, partial dependency cannot occur structurally; the focus here is removing unrelated concerns from shared tables.

Table	Issue / Status	Change Made & Reason
vehicles_1nf — seller data	Seller name, contact, and type are determined by the seller, not by the vehicle.	Extracted to sellers; seller_id FK added to vehicles. Seller attributes must depend only on seller identity.
vehicles_1nf — user identity	User credentials (name, email, phone) depend on the account, not on seller-role attributes.	Extracted to users; user_id FK added to sellers. User identity is independent of seller-specific data such as rating or type.
vehicles_1nf — category data	Category name and description depend on the category, not on the vehicle.	Extracted to categories; category_id FK added to vehicles. Category data is fully determined by category identity alone.
vehicles_1nf — location data	City, state, country depend on the location, not on the vehicle.	Extracted to locations; location_id FK added to vehicles. Location data must not be repeated per vehicle row.
vehicle_images, locations, categories	All non-key attributes depend fully on the single-column surrogate PK.	No change required. 2NF is automatically satisfied with a single-column PK.

4. Third Normal Form (3NF)

Rule: The table is in 2NF and no non-key attribute transitively depends on the primary key (no non-key column depends on another non-key column).

Table	Issue / Status	Change Made & Reason
users — admin permissions	Transitive dependency: user_id → role = 'admin' → permissions.	Extracted to admins with user_id FK. Admin-specific data must not reside in the general users table; this removes the transitive chain.
sellers — rating derived from reviews	Seller rating is an aggregate of review scores and transitively depends on reviews.	Extracted to reviews; sellers.rating kept as a cached aggregate only. Raw review data and derived rating must be separated to prevent update anomalies.
vehicles_2nf — buyer inquiries	Buyer messages are transactions linked to vehicles and sellers — not vehicle attributes.	Extracted to inquiries with FKs to users, vehicles, sellers. Transactions stored as vehicle attributes create transitive dependencies through seller_id.
vehicles_2nf — comparison sessions	Comparison sessions group vehicle IDs through a user context — a user-level event, not a vehicle attribute.	Extracted to comparisons with user_id and up to three vehicle_id FKs. Storing sessions in vehicles creates multi-valued transitive dependencies.
users — saved vehicles	Saved vehicles form a repeating group inside the user row — a many-to-many association.	Extracted to favorites junction table (user_id, vehicle_id). M:N associations require a dedicated junction table.
users — search history	Search events are logs triggered by users — not attributes of the user entity.	Extracted to search_logs with user_id FK. Event logs stored as entity attributes create multi-valued transitive dependencies.
locations, categories, vehicle_images, favorites	All non-key attributes depend directly and solely on the PK. No non-key column depends on another non-key column.	No change required. 3NF is already satisfied.

5. Normalization Summary

Table	1NF	2NF	3NF	Key Action
locations	✓	✓	✓	City and state split from raw flat table (1NF).
categories	✓	✓	✓	Extracted to remove repeated category attributes (2NF).
users	✓	✓	✓	Extracted from sellers to separate user identity from role (2NF).
admins	✓	✓	✓	Extracted from users to remove admin transitive dependency (3NF).
sellers	✓	✓	✓	Extracted from vehicles; seller attributes are not vehicle attributes (2NF).
vehicles	✓	✓	✓	Partial dependencies on seller, location, and category removed (2NF).
vehicle_images	✓	✓	✓	Multi-valued image column extracted to own table (1NF).
favorites	✓	✓	✓	User-vehicle M:N association extracted as junction table (3NF).
comparisons	✓	✓	✓	Comparison events extracted; not vehicle attributes (3NF).
inquiries	✓	✓	✓	Buyer messages extracted; not vehicle or seller attributes (3NF).
reviews	✓	✓	✓	Extracted to remove rating transitive dependency from sellers (3NF).
search_logs	✓	✓	✓	Search events extracted; not user attributes (3NF).

Table 1: Normalization status — all 12 tables (✓ = satisfies the normal form)

6. Final 3NF Schema

PK = Primary Key FK = Foreign Key NN = NOT NULL

6.1 locations

Column	Type	Constraint	PK	FK	Notes
location_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
city	VARCHAR(100)	NOT NULL			City name.
state	VARCHAR(100)	NOT NULL			Province / region.
country	VARCHAR(80)	DEFAULT 'Pakistan'			Country name.

6.2 categories

Column	Type	Constraint	PK	FK	Notes
category_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
category_name	VARCHAR(80)	NOT NULL, UNIQUE			Category label.
description	TEXT	NULL			Optional description.

6.3 users

Column	Type	Constraint	PK	FK	Notes
user_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
name	VARCHAR(120)	NOT NULL			Full display name.
email	VARCHAR(180)	NOT NULL, UNIQUE			Login email.
password_hash	VARCHAR(255)	NOT NULL			Bcrypt hash.
phone	VARCHAR(20)	NULL			Contact number.
role	ENUM	DEFAULT 'buyer'			admin / seller / buyer.
is_active	TINYINT(1)	DEFAULT 1			Soft-delete flag.

6.4 admins

Column	Type	Constraint	PK	FK	Notes
admin_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
user_id	INT UNSIGNED	NOT NULL, UNIQUE		•	FK → users. CASCADE.
permissions	JSON	DEFAULT '{}'			Admin access rights.

6.5 sellers

Column	Type	Constraint	PK	FK	Notes
seller_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
user_id	INT UNSIGNED	NOT NULL, UNIQUE		•	FK → users. CASCADE.
location_id	INT UNSIGNED	NULL		•	FK → locations. SET NULL.
business_name	VARCHAR(150)	NULL			Dealer / showroom name.
rating	DECIMAL(3,2)	DEFAULT 0.00			Cached aggregate from reviews.
seller_type	ENUM	DEFAULT 'individual'			individual / dealer / certified.

6.6 vehicles

Column	Type	Constraint	PK	FK	Notes
vehicle_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
brand	VARCHAR(80)	NOT NULL			Manufacturer name.
model	VARCHAR(120)	NOT NULL			Model name.
variant	VARCHAR(150)	NULL			Trim / variant level.
year	YEAR	CHECK > 1980			Manufacturing year.
price	BIGINT UNSIGNED	CHECK > 0			Listed price (PKR).
mileage	INT UNSIGNED	DEFAULT 0			Odometer reading (km).
fuel_type	ENUM	NOT NULL			Petrol / Diesel / Hybrid / EV / CNG.
transmission	ENUM	NOT NULL			Automatic / Manual / CVT / DCT.
body_type	ENUM	NOT NULL			Sedan / Hatchback / SUV / etc.
engine_cc	INT UNSIGNED	NULL			Engine displacement (cc).
color	VARCHAR(60)	NULL			Exterior colour.
status	ENUM	DEFAULT 'available'			available / sold / featured.
category_id	INT UNSIGNED	NULL		•	FK → categories. SET NULL.
seller_id	INT UNSIGNED	NOT NULL		•	FK → sellers. CASCADE.
location_id	INT UNSIGNED	NULL		•	FK → locations. SET NULL.

6.7 vehicle_images

Column	Type	Constraint	PK	FK	Notes
image_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
vehicle_id	INT UNSIGNED	NOT NULL		•	FK → vehicles. CASCADE.
image_url	VARCHAR(400)	NOT NULL			CDN image URL.
is_primary	TINYINT(1)	DEFAULT 0			Cover image flag.

6.8 favorites

Column	Type	Constraint	PK	FK	Notes
favorite_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
user_id	INT UNSIGNED	NOT NULL		•	FK → users. CASCADE.
vehicle_id	INT UNSIGNED	NOT NULL		•	FK → vehicles. CASCADE. UNIQUE with user_id.

6.9 comparisons

Column	Type	Constraint	PK	FK	Notes
comparison_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
user_id	INT UNSIGNED	NOT NULL		•	FK → users. CASCADE.
vehicle_id_1	INT UNSIGNED	NOT NULL		•	First vehicle. CASCADE.
vehicle_id_2	INT UNSIGNED	NOT NULL		•	Second vehicle. CASCADE.
vehicle_id_3	INT UNSIGNED	NULL		•	Third vehicle (optional). SET NULL.

6.10 inquiries

Column	Type	Constraint	PK	FK	Notes
inquiry_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
user_id	INT UNSIGNED	NULL		•	FK → users. SET NULL (guest support).
vehicle_id	INT UNSIGNED	NOT NULL		•	FK → vehicles. CASCADE.
seller_id	INT UNSIGNED	NOT NULL		•	FK → sellers. CASCADE.
message	TEXT	NOT NULL			Inquiry message body.
status	ENUM	DEFAULT 'new'			new / read / replied / closed.

6.11 reviews

Column	Type	Constraint	PK	FK	Notes
review_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
reviewer_id	INT UNSIGNED	NOT NULL		•	FK → users. CASCADE.
seller_id	INT UNSIGNED	NOT NULL		•	FK → sellers. CASCADE.
rating	TINYINT UNSIGNED	CHECK (1–5)			Star rating.
comment	TEXT	NULL			Review text.

6.12 search_logs

Column	Type	Constraint	PK	FK	Notes
log_id	INT UNSIGNED	AUTO_INC, NN	•		Primary key.
user_id	INT UNSIGNED	NULL		•	FK → users. No cascade.
query_text	VARCHAR(300)	NULL			Search keywords.
filters	JSON	NULL			Serialised active filters.
results_count	INT UNSIGNED	NULL			Results returned.